

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$\mathcal{K}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\bullet 1\bullet 2\bullet 3}$, {●1→ -a, ●2→ -b, ●3→ -c}, StrongGenSet[{2, 3}, GenSet[-(2,3)]]]	$\mathcal{K}^{\#1}_{0^{-}\alpha\beta\chi} - \frac{1}{2} \eta_{\alpha\chi} \mathcal{K}^{\#1}_{1^{-}\beta} + \frac{1}{2} \eta_{\alpha\beta} \mathcal{K}^{\#1}_{1^{-}\chi} + \frac{4}{3} \mathcal{K}^{\#1}_{2^{-}\beta\chi\alpha} + \mathcal{K}^{\#2}_{1^{+}\beta\chi} n_{\alpha} + \frac{1}{3} \eta_{\alpha\chi} \mathcal{K}^{\#1}_{0^{+}} n_{\beta} - \mathcal{K}^{\#1}_{1^{+}\alpha\chi} n_{\beta} + \mathcal{K}^{\#1}_{2^{+}\alpha\chi} n_{\beta} -$ $\frac{1}{2} \mathcal{K}^{\#1}_{1^{-}\chi} n_{\alpha} n_{\beta} - \mathcal{K}^{\#2}_{1^{-}\chi} n_{\alpha} n_{\beta} - \frac{1}{3} \eta_{\alpha\beta} \mathcal{K}^{\#1}_{0^{+}} n_{\chi} + \mathcal{K}^{\#1}_{1^{+}\alpha\beta} n_{\chi} - \mathcal{K}^{\#1}_{2^{+}\alpha\beta} n_{\chi} + \frac{1}{2} \mathcal{K}^{\#1}_{1^{-}\beta} n_{\alpha} n_{\chi} + \mathcal{K}^{\#2}_{1^{-}\beta} n_{\alpha} n_{\chi}$	$\mathcal{I}_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$\mathcal{K}^{\#1}_{0^{+}}$	Symmetry[0, $\mathcal{K}^{\#1}_{0^{+}}$, {}, StrongGenSet[{}, GenSet[]]]	$\mathcal{K}^{\beta}_{\alpha\beta} n^{\alpha}$	$\mathcal{I}^{\#1}_{0^{+}}$
$\mathcal{K}^{\#1}_{0^{-}\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\#1\bullet 1\bullet 2\bullet 3}$, {●1→ -a, ●2→ -b, ●3→ -c}, StrongGenSet[{1, 2, 3}, GenSet[-(1,2), -(2,3)]]]	$\frac{1}{3} \mathcal{K}_{\alpha\beta\chi} - \frac{1}{3} \mathcal{K}_{\beta\alpha\chi} + \frac{1}{3} \mathcal{K}_{\chi\alpha\beta} - \frac{1}{3} \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{3} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\delta} - \frac{1}{3} \mathcal{K}_{\delta\beta\chi} n_{\alpha} n^{\delta} +$ $\frac{1}{3} \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\delta} + \frac{1}{3} \mathcal{K}_{\delta\alpha\chi} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{3} \mathcal{K}_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{3} \mathcal{K}_{\delta\alpha\beta} n_{\chi} n^{\delta}$	$\mathcal{I}^{\#1}_{0^{-}\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{1^{+}\alpha\beta}$	Symmetry[2, $\mathcal{K}^{\#1\bullet 1\bullet 2}$, {●1→ -a, ●2→ -b}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\frac{1}{2} \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\beta\alpha\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}^{\#1}_{1^{+}\alpha\beta}$
$\mathcal{K}^{\#2}_{1^{+}\alpha\beta}$	Symmetry[2, $\mathcal{K}^{\#2\bullet 1\bullet 2}$, {●1→ -a, ●2→ -b}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\mathcal{K}_{\chi\alpha\beta} n^{\chi} + \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} - \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}^{\#2}_{1^{+}\alpha\beta}$
$\mathcal{K}^{\#1}_{1^{-}\alpha}$	Symmetry[1, $\mathcal{K}^{\#1\bullet 1}$, {●1→ -a}, StrongGenSet[{}, GenSet[]]]	$-\mathcal{K}^{\beta}_{\alpha\beta} + \mathcal{K}^{\chi}_{\beta\chi} n_{\alpha} n^{\beta} + \mathcal{K}_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$\mathcal{I}^{\#1}_{1^{-}\alpha}$
$\mathcal{K}^{\#2}_{1^{-}\alpha}$	Symmetry[1, $\mathcal{K}^{\#2\bullet 1}$, {●1→ -a}, StrongGenSet[{}, GenSet[]]]	$\mathcal{K}_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$\mathcal{I}^{\#2}_{1^{-}\alpha}$
$\mathcal{K}^{\#1}_{2^{+}\alpha\beta}$	Symmetry[2, $\mathcal{K}^{\#1\bullet 1\bullet 2}$, {●1→ -a, ●2→ -b}, StrongGenSet[{1, 2}, GenSet[(1,2)]]]	$-\frac{1}{2} \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\beta\alpha\chi} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} \mathcal{K}^{\delta}_{\chi\delta} n^{\chi} + \frac{1}{3} \mathcal{K}^{\delta}_{\chi\delta} n_{\alpha} n_{\beta} n^{\chi} + \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}^{\#1}_{2^{+}\alpha\beta}$
$\mathcal{K}^{\#1}_{2^{-}\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\#1\bullet 1\bullet 2\bullet 3}$, {●1→ -a, ●2→ -b, ●3→ -c}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$-\frac{1}{4} \mathcal{K}_{\alpha\beta\chi} + \frac{1}{4} \mathcal{K}_{\beta\alpha\chi} + \frac{1}{2} \mathcal{K}_{\chi\alpha\beta} - \frac{3}{8} \eta_{\beta\chi} \mathcal{K}^{\delta}_{\alpha\delta} + \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}^{\delta}_{\beta\delta} - \frac{3}{8} \mathcal{K}^{\delta}_{\beta\delta} n_{\alpha} n_{\chi} + \frac{3}{8} \mathcal{K}^{\delta}_{\alpha\delta} n_{\beta} n_{\chi} + \frac{1}{4} \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\delta} +$ $\frac{1}{4} \mathcal{K}_{\delta\beta\chi} n_{\alpha} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{K}^{\epsilon}_{\delta\epsilon} n_{\alpha} n^{\delta} - \frac{1}{4} \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\delta} - \frac{1}{4} \mathcal{K}_{\delta\alpha\chi} n_{\beta} n^{\delta} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}^{\epsilon}_{\delta\epsilon} n_{\beta} n^{\delta} + \frac{1}{4} \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} -$ $\frac{1}{4} \mathcal{K}_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{2} \mathcal{K}_{\delta\alpha\beta} n_{\chi} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{K}_{\delta\alpha\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}_{\delta\beta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \mathcal{K}_{\delta\beta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{3}{8} \mathcal{K}_{\delta\alpha\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon}$	$\mathcal{I}^{\#1}_{2^{-}\alpha\beta\chi}$